# EXECUTIVE SUMMARY

## From Illustrative to Measured: Task-Oriented Semantic mIoT over NTN for Remote Agriculture and Environmental Sensing

Rao Yenamandra (DigitalTwinSim, San Diego, CA, USA) · Mubanga Nsofu (Vodacom, Johannesburg, South Africa) · Ram Asokan (Wireless Research Center of North Carolina, USA)

Target: SPACOMM 2027 (IARIA), Lisbon, April 18–22, 2027 · Submission deadline: December 28, 2026 · Status: manuscript complete (8 pp.), not submitted

### THE PROBLEM

An earlier IEEE submission of this work was rejected: reviewers found no substantive experiments — the headline 32.5× gain was an illustrative calculation, a section titled “Required Results” made the paper look unfinished, five applications were too broad, and template/proofreading defects hurt credibility. This revision answers every criticism with measured evidence.

### WHAT WAS DONE

The paper was rewritten from scratch around a three-stage, trace-driven simulation (seeded, reproducible, 5 held-out evaluation seeds) comparing semantic/task-oriented reporting against raw periodic, compressed periodic, and event-triggered baselines. Scope was narrowed to two India-oriented applications: (1) irrigation decisions for smart agriculture (soil-moisture traces, 6-hour deadline) and (2) flood warnings for environmental monitoring (river-level traces, 45-minute deadline). Link and energy models are anchored in published NTN field measurements, and the architecture figure was completely redesigned.

### HEADLINE RESULTS (all measured, mean ± sd over 5 seeds)

- Report suppression dominates: of the 42.6× offered-byte gain, 18.9× comes from sending fewer reports (only ~5% admitted) and just 2.25× from smaller payloads. The 60 B header caps the airtime gain at 19.6×.
- Capacity: at 95% deadline reliability, semantic reporting supports 1,600 devices/beam for irrigation vs 50 for raw periodic. For flood warnings, raw and compressed reporting collapse entirely (0 supportable devices); semantic supports 400.
- Task quality: flood-warning recall 0.928 (semantic) vs 0.157 (raw, which misses 84% of floods) vs 0.657 (event-triggered). Task-oriented gain: 104.6× (floods), 19.6× (irrigation).
- Energy: ~300 J vs ~28 kJ per device per 90 days (≈99% saving); the honest uncongested figure is ~95%, since part of raw's cost is a congestion retry-storm.
- Adaptation: a drift detector plus one threshold update restores flood-warning recall from 0.746 (stale dry-season tuning) to 1.000.

### NEXT STEPS (framed as future work in the paper)

Validation on field sensor datasets (rainfall, soil-moisture, river-level records); multi-constellation link sweeps; NTN-specific random-access (NPRACH) modeling and hardware energy measurements on MCU-class devices; heterogeneous device/receiver decision rules; alternative deadline/reliability operating points. All reported gains are resource and task-outcome gains bounded to the evaluated applications — no Shannon-capacity claims.

### REMAINING STEPS

1) Confirm affiliations as listed above. 2) Decide: submit at 8 pages (2 pages over the included 6 — extra-page fees apply) or cut to 6. 3) Sanity-check the six flagged simulation assumptions. 4) Submit via the SPACOMM 2027 system before December 28, 2026.